

Smart Home IoT Control System

Project Report

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Student Work Archive System
Sample Document for Testing Purposes

Abstract

This project implements a comprehensive Internet of Things (IoT) solution for smart home automation. The system enables users to control various home appliances including lighting, temperature, and security systems through a unified mobile application.

The technical implementation utilizes ESP32 microcontrollers as edge devices, MQTT protocol for reliable message passing, and Flutter for cross-platform mobile development. The system supports both local and cloud-based control modes.

Key features include:

- Voice control integration with Google Assistant
- Automated scheduling and routines
- Energy consumption monitoring
- Motion-triggered security alerts
- Multi-user access with role-based permissions

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Chapter 1

Introduction

Smart home technology has become increasingly accessible to consumers. This project aims to create an affordable yet feature-rich home automation system that can be easily deployed and configured by non-technical users.

This section contains additional technical details and supporting information relevant to the chapter topic. The content demonstrates the depth of research and analysis conducted throughout this study.

Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 2

System Architecture

The system follows a hub-and-spoke architecture with a Raspberry Pi 4 serving as the central hub. ESP32 microcontrollers act as smart nodes controlling individual devices. Communication between nodes uses MQTT over WiFi.

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Chapter 3

Hardware Design

Each smart node consists of an ESP32 development board, relay modules for switching, and appropriate sensors (temperature, humidity, motion). PCB designs were created using KiCad for a more compact final product.

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Chapter 4

Software Implementation

The mobile app was developed using Flutter for cross-platform compatibility. The backend uses Node.js with Express, and MongoDB for data persistence. Real-time updates are handled through WebSocket connections.

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Chapter 5

Testing and Results

The system was tested in a real home environment for 2 months. Response time averaged 200ms for local commands and 500ms for cloud commands. Energy monitoring showed potential savings of 15-20% through automated scheduling.

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Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 6

Conclusion

The Smart Home IoT Control System provides a practical and cost-effective solution for home automation. The modular design allows for easy expansion and customization based on user needs.

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Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

References

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